Comparison between the Additive and Substitutive Models

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ABSTRACT

Several different algorithms have been suggested for matched target detection for solid and gas point targets: theoretically, gas targets lead to an additive point target detection algorithm while solid targets lead to a substitutive point target detection algorithm. In this paper, we consider deviations that may occur in real life non-stationary backgrounds that would complicate the choice of algorithm.

In particular, we have identified three possible weaknesses to the substitutive algorithm. First, incorrect estimates of the background for a minority of pixels can lead to considerably worse results in the substitutive algorithm compared to the additive algorithm. Second, an accurate knowledge of the amplitude (or power) of the target is needed for the substitutive algorithm compared to the additive algorithm. Thirdly, due to adjacency effects, it is not obvious which model to use in certain cases.

For this latter case, we consider a hybrid model originally proposed by A. Ziemann et al. [1]. This model, with an adaptive parameter estimate of the form of the detection to use, appears to be a reasonable substitute for the two aforementioned algorithms.

Keywords: Hyperspectral, Target detection, Simplex.

1. Introduction

1.1. Background

Hyperspectral images are taken with dozens, even hundreds, of contiguous wavelength bands from across the electromagnetic spectrum [1]. The Hyperspectral image is in the shape of a three-dimensional cube, where the third dimension represents the spectral bands. Each pixel is a vector whose components are the relative light's intensity received by the sensor for a specific wavelength.

Processing hyperspectral images has many applications, ranging from agriculture, military, and environment to the food industry.

1.2. Target Detection in Hyperspectral Images

For a broad spectrum of wavelengths, each material has its unique spectral representation – its **spectral signature**. Target detection in hyperspectral images is based on either the target's signature or a linear combination of different elements that the target is composed of, such as water, soil, etc. The algorithms use these signatures as "fingerprints" to locate the target in the image.

Detection algorithms are based on the target model, which is either additive (for transparent targets) or substitutive (for opaque ones). The algorithms at the center of this article are the Additive AMF, the Substitutive AMF, and the Simplex AMF. The first two were uniquely formulated for the additive and substitutive target models, and are supposed, theoretically, to deliver the optimal results vis a vis the signal to noise ratio. However, these algorithms, and the substitutive AMF especially, are highly sensitive to the degree of accuracy in their input.

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1.3. Study Objectives

In our study, we compared the three algorithms' outputs for various inputs. Our main objective was to determine the optimal model of the Additive and Substitutive AMF. Our assumption was that the target is a Sub-Pixel Point Target, meaning that it only influences isolated pixels, rather than a group of pixels.

We show in this paper that when there is a detection error, caused by inaccurate background estimation, the Substitutive AMF yields a greater error than the Additive AMF. However, when using a specific subspace, the Simplex can perform as the alternate algorithm, meaning it will give reliable results. And finally, when there is an error in the target's amplitude, the Additive AMF is the most fitting algorithm.

2. Methodology

The general agreement is that the target is embedded in the background in one of the two implantation models:

1. Additive Model - a gas matter integrates with the background; hence the target implantation is a linear addition of its energy to that of the pixel.
2. Substitutive Model - a solid matter covers part of the background; hence the target implantation is a replacement of a portion of the pixel's energy with that of the target.

For both implantation models, the target detection algorithms are derived [2] from the RX filter [3], an anomaly detection algorithm that is applied when the target's signature is unknown, as given in Eq. 1:

$$RX = (\boldsymbol{x}' - \boldsymbol{m})^T \bar{\bar{\phi}}^{-1} (\boldsymbol{x}' - \boldsymbol{m}) \tag{1}$$

Where $\boldsymbol{x}'$ is the pixel under examination, $\boldsymbol{m}$ is the background estimation, computed from the average of the pixel's eight closest neighbors, and $\bar{\bar{\phi}}$ is the covariance matrix, estimated from the background data.

The RX filter searches for irregularities in the pixels, where the potential of a target location is higher.

Let us assume that the target is additive. Then the implantation is of the form:

$$\boldsymbol{x}' = \boldsymbol{x} + p\boldsymbol{t} \tag{2}$$

Where $\boldsymbol{x}$ is the background pixel, $\boldsymbol{t}$ is the target's signature, p is the target's intensity, satisfying $p \in [0,1]$.
If $\boldsymbol{m}$ is a correct estimation of $\boldsymbol{x}$, then we get:

$$\begin{aligned} \boldsymbol{x}' - \boldsymbol{m} &= \boldsymbol{x} + p\boldsymbol{t} - \boldsymbol{m} \\ &\approx p\boldsymbol{t} \end{aligned} \tag{3}$$

Combining equations (1) and (3) will produce:

$$RX(\boldsymbol{x}') = \boldsymbol{t}^T \bar{\bar{\phi}}^{-1} (\boldsymbol{x}' - \boldsymbol{m}) \tag{4}$$

Notice that we neglect p because multiplication by a constant does not affect the algorithm's output.
Now, let us normalize the target's signature and obtain the AMF algorithm for an additive target:

$$AMF_{additve} = \frac{\boldsymbol{t}^T \bar{\bar{\phi}}^{-1} (\boldsymbol{x}' - \boldsymbol{m})}{\sqrt{\boldsymbol{t}^T \bar{\bar{\phi}}^{-1} \boldsymbol{t}}} \tag{5}$$

Alternatively, let us assume a substitutive target, that is:

$$\boldsymbol{x}' = (1-p)\boldsymbol{x} + p\boldsymbol{t} \tag{6}$$

Again, under the assumption that $\boldsymbol{m}$ is a correct estimation of $\boldsymbol{x}$:

$$\begin{aligned} \boldsymbol{x}' - \boldsymbol{m} &= (1-p)\boldsymbol{x} + p\boldsymbol{t} - \boldsymbol{m} \\ &\approx p(\boldsymbol{t} - \boldsymbol{m}) \end{aligned} \tag{7}$$

Accordingly, Eq. (1) and (7) will produce:

$$RX(\boldsymbol{x}') = (\boldsymbol{t} - \boldsymbol{m})^T \bar{\bar{\phi}}^{-1} (\boldsymbol{x}' - \boldsymbol{m}) \tag{8}$$

After normalization, the obtained AMF algorithm for a substitutive target is:

$$AMF_{substitutive} = \frac{(\boldsymbol{t} - \boldsymbol{m})^T \bar{\bar{\phi}}^{-1} (\boldsymbol{x}' - \boldsymbol{m})}{\sqrt{(\boldsymbol{t} - \boldsymbol{m})^T \bar{\bar{\phi}}^{-1} (\boldsymbol{t} - \boldsymbol{m})}} \tag{9}$$

In brief, the algorithms compare the whitened target's spectrum to the whitened proposed pixel and set the score by their level of correlation.

These two algorithms depend extensively on an accurate estimation of the image background. However, from Eq. 9, we notice that the substitutive AMF has a second-order dependency of the background estimation, therefore any error in the background estimation causes an even bigger error in the detection.

The last algorithm we will discuss is the Simplex AMF algorithm. AMF is a Multiple-Input Target algorithm [4], whose premise is that the target is a linear, non-negative, combination of the signatures of elemental materials called Endmembers. The Algorithm, presented by Ziemann and Theiler, is defined as such:

$$\boldsymbol{a}_{\boldsymbol{x}'} = argmin_{\boldsymbol{a}} \left\| \boldsymbol{x}' - \boldsymbol{m} - \bar{\bar{E}} \boldsymbol{a} \right\| \ \text{ s.t. } \ \boldsymbol{a} \geq \boldsymbol{0} \tag{10}$$

$$D(\boldsymbol{x}') = \left\| \bar{\bar{\phi}}^{-1/2} \bar{\bar{E}} \boldsymbol{a}_{\boldsymbol{x}'} \right\| \tag{11}$$

Where $\bar{\bar{E}}$ is the endmembers matrix, whitened by $\bar{\bar{\phi}}^{-1/2}$.
Geometrically, Eq. 11 is equivalent to:

$$D(\boldsymbol{x}') = \left(\bar{\bar{E}} \boldsymbol{a}_{\boldsymbol{x}'}\right)^T \bar{\bar{\phi}}^{-1} (\boldsymbol{x}' - \boldsymbol{m}) \tag{12}$$

After normalization, the simplex AMF is obtained by:

$$AMF_{simplex} = \frac{\left(\bar{\bar{E}} \boldsymbol{a}_{\boldsymbol{x}'}\right)^T \bar{\bar{\phi}}^{-1} (\boldsymbol{x}' - \boldsymbol{m})}{\sqrt{\left(\bar{\bar{E}} \boldsymbol{a}_{\boldsymbol{x}'}\right)^T \bar{\bar{\phi}}^{-1} \left(\bar{\bar{E}} \boldsymbol{a}_{\boldsymbol{x}'}\right)}} \tag{13}$$

The subspace we chose to work with is:

$$\bar{\bar{E}} = [\boldsymbol{t}, \boldsymbol{t} - \boldsymbol{m}] \ , \tag{14}$$

with the reason being that the projected pixel will be somewhat a compromise between the additive and the substitutive models.

Overall, the formulas presented in Eq. 5 and 9 are theoretically ideal for additive and substitutive target detection, respectively, but only under the assumption that the algorithms' inputs are all prior knowledge. In this paper, we examine the performances of the algorithms in cases in which prior knowledge of the target signature is inaccurate or unknown. More precisely - when the target's amplitude, the target's model, or the background estimation is uncertain.

3. Comparison Method

Our method of comparing the algorithms' performances is the Receiver Operating Characteristic (ROC curve), and the Area Under Curve (AUC) [5]. The ROC curve is a graphical plot, illustrating the probability of detection as a function of the false alarm rate. It is generated as follows:

1. First, you form the algorithm's output in target-absent and target-present histograms.
2. Next, you normalize the histograms so that the cumulative distribution function (CDF) will have a maximum value of one.
3. Finally, you sweep a threshold variable, th, through the histograms, and calculate the inverse CDF, as the area that is under the curve and on the right to th. The target-present value is P_D, and the target-absent value is P_{FA}, thus forming two vectors of detection probabilities that we can then plot in pairs.

A diagram for the testing process is shown below:

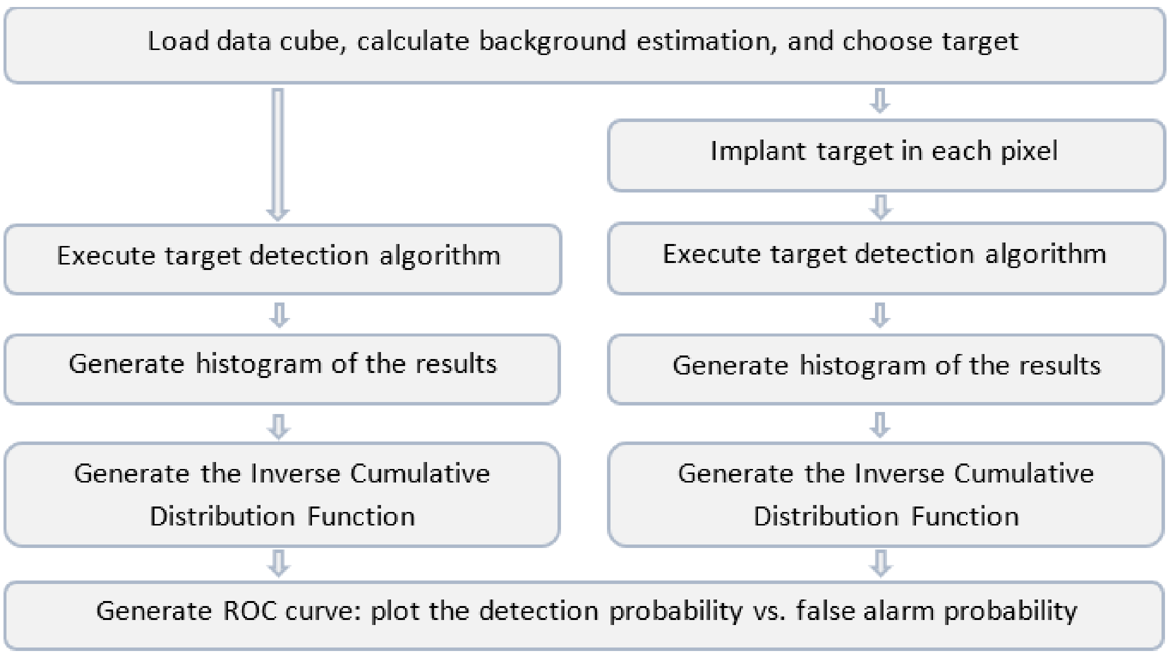


Figure 1: Testing process diagram

Ideally, a ROC curve should resemble the step function, denoting an algorithm that optimally detects targets, meaning $P_D = 1 \, \forall P_{FA}$. On the other hand, a ROC curve that resembles a linear function ($P_{FA} = P_D$) will indicate an algorithm that is indifferent to targets, and therefore ineffective.

In most real-life applications, the weights of detection probability and false alarm rate are not the same. Therefore, the x-axis of the ROC curve is not from 0 to 1, but rather from 0 to a specific threshold, th, where $th \in [0,1]$.
The AUC (Area Under the Curve) metric is derived from the ROC curve as such:

$$A_{th} = \frac{A - \frac{1}{2}(th)^2}{th - \frac{1}{2}(th)^2} \tag{15}$$

Where A is the calculated area under the ROC curve for a fixed threshold. A_{th} is then normalized so that it yields zero when $P_{FA} = P_D$ and one when the ROC curve is ideal, regardless of the threshold we choose.

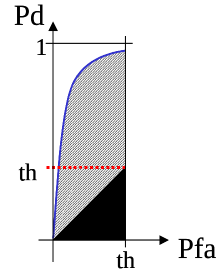


Figure 2: Illustration of the ROC Curve

4. Data Sets

The Data Cubes used for analysis in this article are:

1. RIT Blind test Dataset - a hyperspectral image of Cooke City, Montana, collected by Rochester Institute of Technology (RIT) [5]. It consists of 126 spectral bands, and its spatial dimension is 280 by 800 pixels.
2. Viareggio Dataset - a hyperspectral image of Viareggio, Tuscany, collected by CISAM [6]. It consists of 512 spectral bands, and its spatial dimension is 375 by 450 pixels.

The RIT dataset was used for the prime analysis, and the Viareggio dataset was used for verification of the results.

Throughout the study, we evaluated the algorithms' performances using six different target signatures. Five of which were the mean of the various image segments, and the sixth one was the global mean of the pixels.

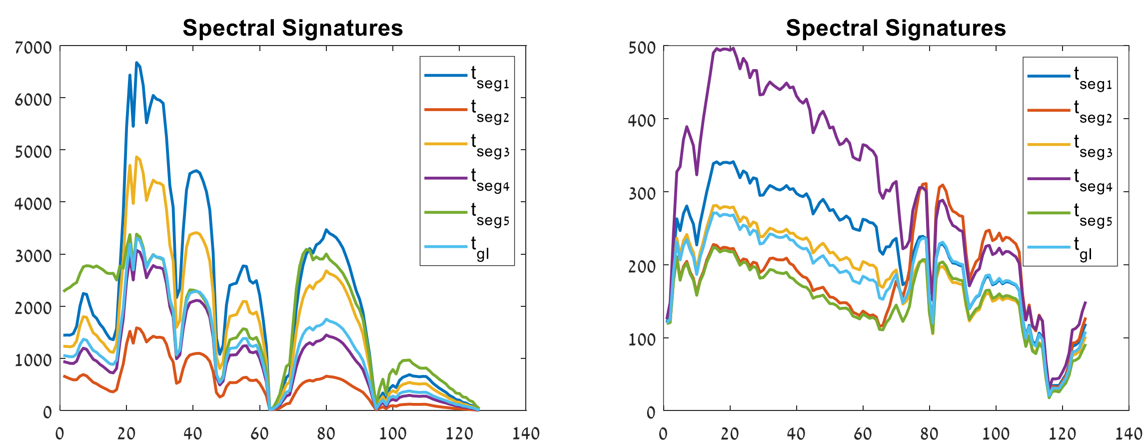


Figure 1: Target's spectrum, for the RIT dataset (left) and the Viareggio dataset (right)

5. Experiments

In the following sub-sections, we will list the various tests we conducted.

5.1. False Alarms

As mentioned, our prime goal was to compare the additive and substitutive models, get a better understanding of the preferred detection model, and whether previous studies were right to prefer the additive model. To this end, we activated the two algorithms on the original image, without the implanted target, to compare the false detection results of both, as can be seen in Eq. 16.

$$Diff = |fa_{additive}| - |fa_{substitutive}| \tag{16}$$

5.2. Accurate Knowledge of Target

The next phase of the study was to compare the results of the three algorithms when used with no input errors. Our goal was to assess whether there is a notable advantage to using the Substitutive Algorithm when the target is substitutive and our prior knowledge of the target is accurate, and in which cases the Simplex Algorithm is preferable to the other algorithms.

Our method was to implant the target in the image, using each of the target's models, perform the detection using all three algorithms, and compare the results using the ROC curves.

5.3. Incorrect Amplitude

After conducting a comparison for accurate prior knowledge, we proceeded with a comparison for inaccurate one, i.e., where the target signature's amplitude is unknown. For this purpose, we implanted a target as follows: we multiplied the target by ratio $R > 1$ and the target's intensity by the inverse ratio $1/R$, so that their product is $p\boldsymbol{t}$, as can be seen in Eq. 17.

$$\tilde{p}\tilde{\boldsymbol{t}} = \frac{1}{R}p \cdot R\boldsymbol{t} = p\boldsymbol{t} \tag{17}$$

The substitutive implantation formula is shown in Eq. 18, while the additive one does not change.

$$\boldsymbol{x}' = \left(1 - \frac{1}{R}p\right)\boldsymbol{x} + p\boldsymbol{t} \tag{18}$$

Then, the detection was performed through the three algorithms for a target signature T, and the results were evaluated by the ROC curves. When this is the case, now $\tilde{\boldsymbol{t}} \gg \boldsymbol{m}$, and therefore the effect of an inaccurate background estimation may be reduced.

6. Results

6.1. False Alarm

The following histogram presents the results for the false alarm comparison.

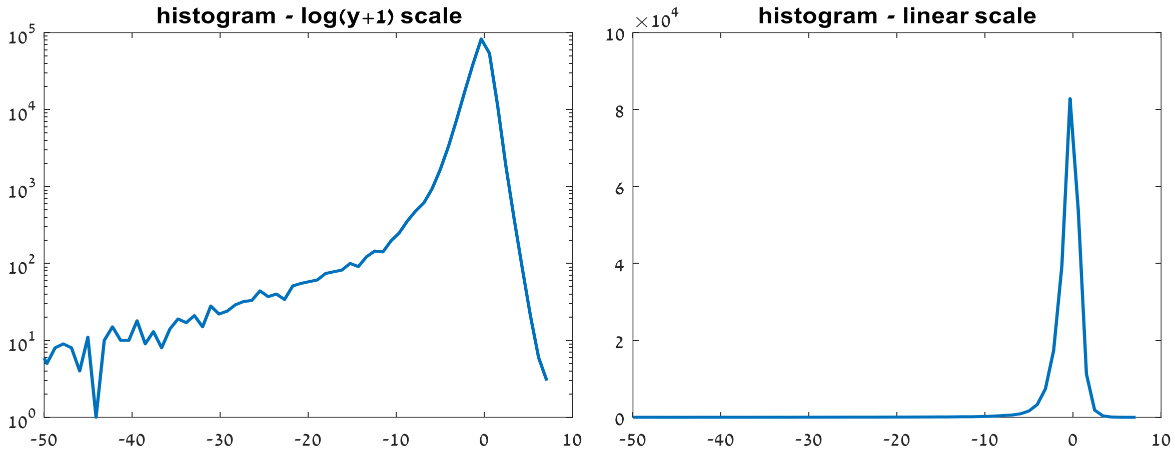
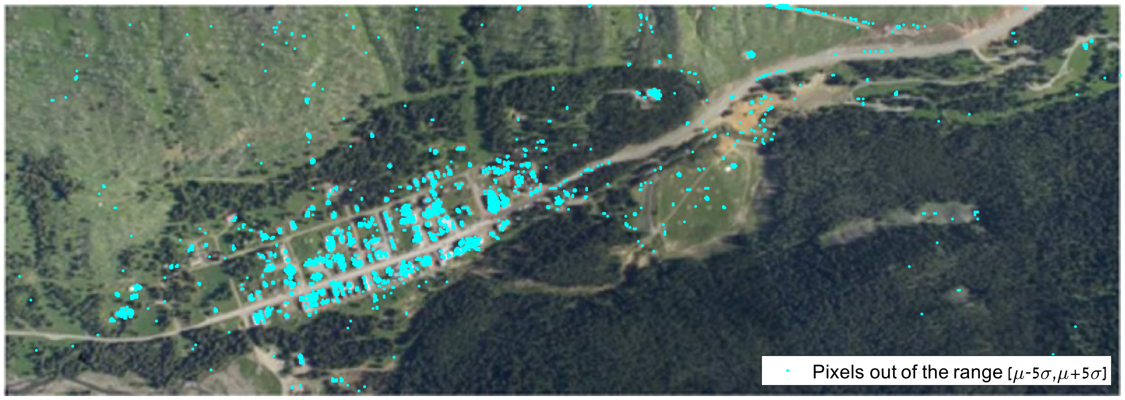


Figure 2: Histogram visualizing the difference between the Additive AMF and the Substitutive AMF. The function used is

$$Diff = |fa_{additive}| - |fa_{substitutive}|$$

It is apparent that the histogram is asymmetrical and has a long tail toward the negative axis, meaning that while most of the pixels get a similar detection error for the two algorithms, a small number of pixels gets a larger error when run through the Substitutive AMF than the Additive AMF. The calculated mean of the results (amongst the highest 99%) is -1.3±0.3, and the standard deviation is 1.4±0.1.

To identify the nature of the pixels that make up the tail of the histogram, we marked these pixels on the images (Fig. 5). We can see that in the RIT image, most of the highlighted pixels are within the urban area, where there is the least correlation between the adjacent pixels. In the Viareggio image, this can be seen more prominently, as the pixels are on the edges between two neighboring segments. We concluded that the difference between the outcomes of the algorithms is indeed due to an inaccurate background estimation.



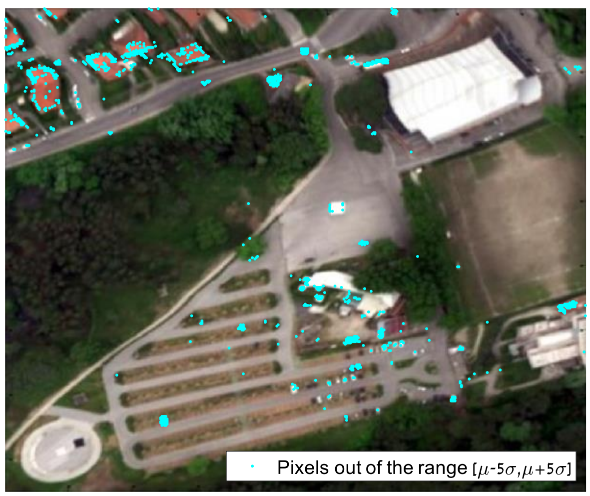


Figure 5: visual of the problematic pixels in the RIT dataset (top image) and in the Viareggio dataset (bottom image)

6.2. Accurate Knowledge of Target

An example of the difference in the histograms of the algorithm's output is demonstrated for the RIT and Viareggio images in Fig. 6.

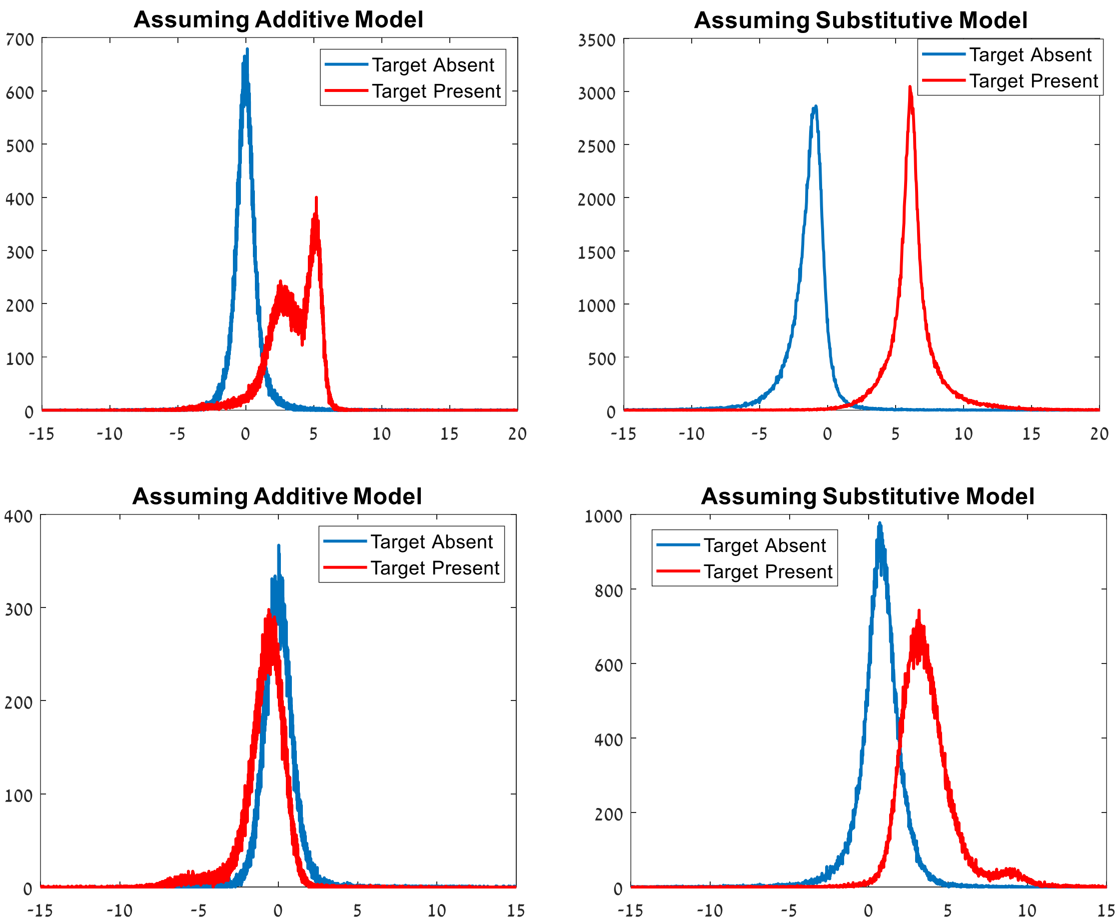


Figure 3: Detection Histograms when input is accurate, and target is substitutive (p=20%), for the RIT image (top row) and for the Viareggio image (bottom row).

In this example, a substitutive target was implanted in the image. The greater the distance between the target-present and target-absent histograms, the better the performance of the algorithm, and in this case the histograms show an advantage to the substitutive AMF over the additive AMF.

Figure 7 shows the ROC curves obtained for each image, assuming accurate prior knowledge of the target.

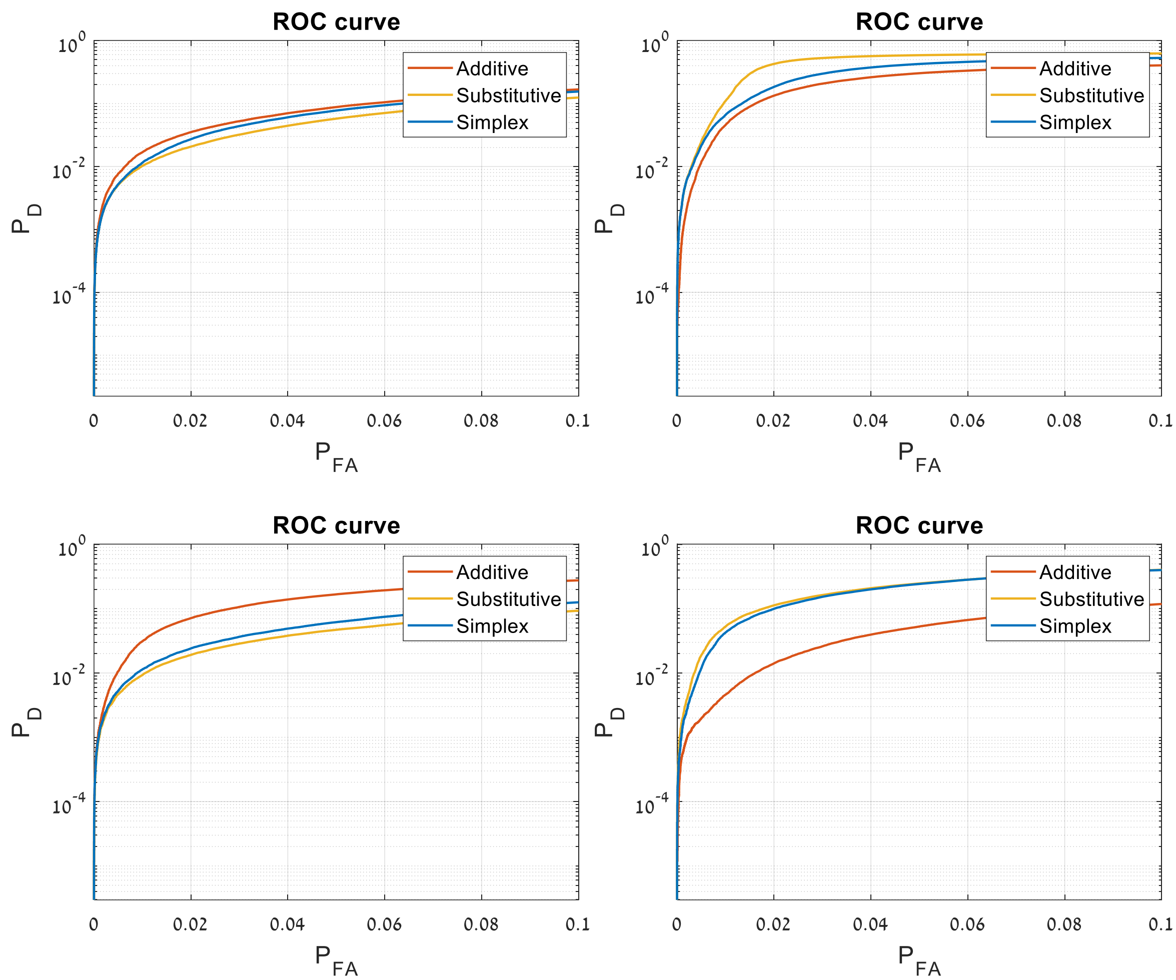


Figure 4: ROC curves for Detection while input is accurate for the RIT dataset with an additive target (top left), RIT dataset with a substitutive target (top right), Viareggio dataset with an additive target (bottom left), and the Viareggio dataset with a substitutive target (bottom right)

As can be seen here, for an additive target, the additive AMF is, as theorized, the preferred algorithm. The same can be said for a substitutive target and the substitutive AMF. Additionally, the Simplex AMF is an intermediate solution between the Additive and the Substitutive AMFs. Therefore, it can be used when the target model is unknown, in order to achieve reliable results.

The AUC numbers shown in Table 1 confirm the results shown in figure 7. For an additive target model, the AUC results of the Additive AMF are the highest ones, and vice versa. Moreover, the AUC results of the Simplex AMF are always in between the results of the other algorithms.

Table 1: AUC calculations, for different scenarios, assuming accurate and inaccurate target model

		RIT Dataset			Viareggio Dataset		
		Additive Detection	Substitutive Detection	Simplex Detection	Additive Detection	Substitutive Detection	Simplex Detection
Additive Target Model	Th=0.1	0.0439	0.0124	0.0326	0.1288	-0.0018	0.0155
	Th=0.01	0.0030	0.0001	0.0004	0.0076	-0.0003	0.0004
Substitutive Target Model	Th=0.1	0.2724	0.6571	0.4214	0.0066	0.2301	0.2233
	Th=0.01	0.0108	0.0315	0.0214	-0.0028	0.0168	0.0106

Nevertheless, the results were not conclusive. Out of the six target signatures we have implanted and detected, one gave mixed results. For this signature, the additive algorithm outperforms the substitutive one for a certain threshold's range, as can be seen in Fig. 8.

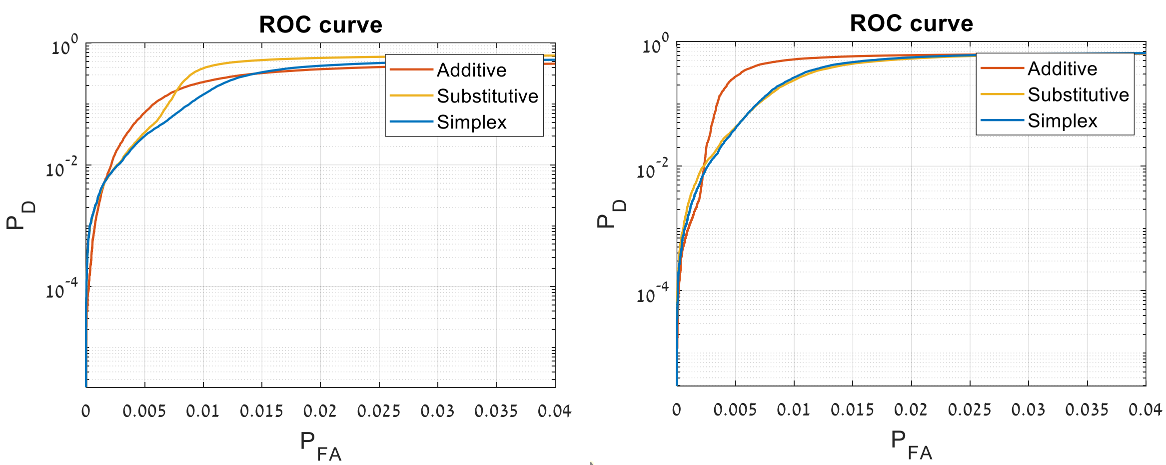


Figure 5: ROC curve which is attributed to the target signal which showed different results when using the substitutive target model in the RIT dataset (left), and the Viareggio dataset (right)

The signatures in question are shown in Fig. 3 as $\boldsymbol{t}_{seg1}$ and $\boldsymbol{t}_{seg4}$ - the mean of the first segment in the RIT dataset and the fourth segment in the Viareggio dataset respectively. These signatures have a higher amplitude than the others in the image.

6.3. Incorrect Amplitude

An example of the difference in the histograms of the algorithm's output is demonstrated for the RIT and Viareggio images in Fig. 9.

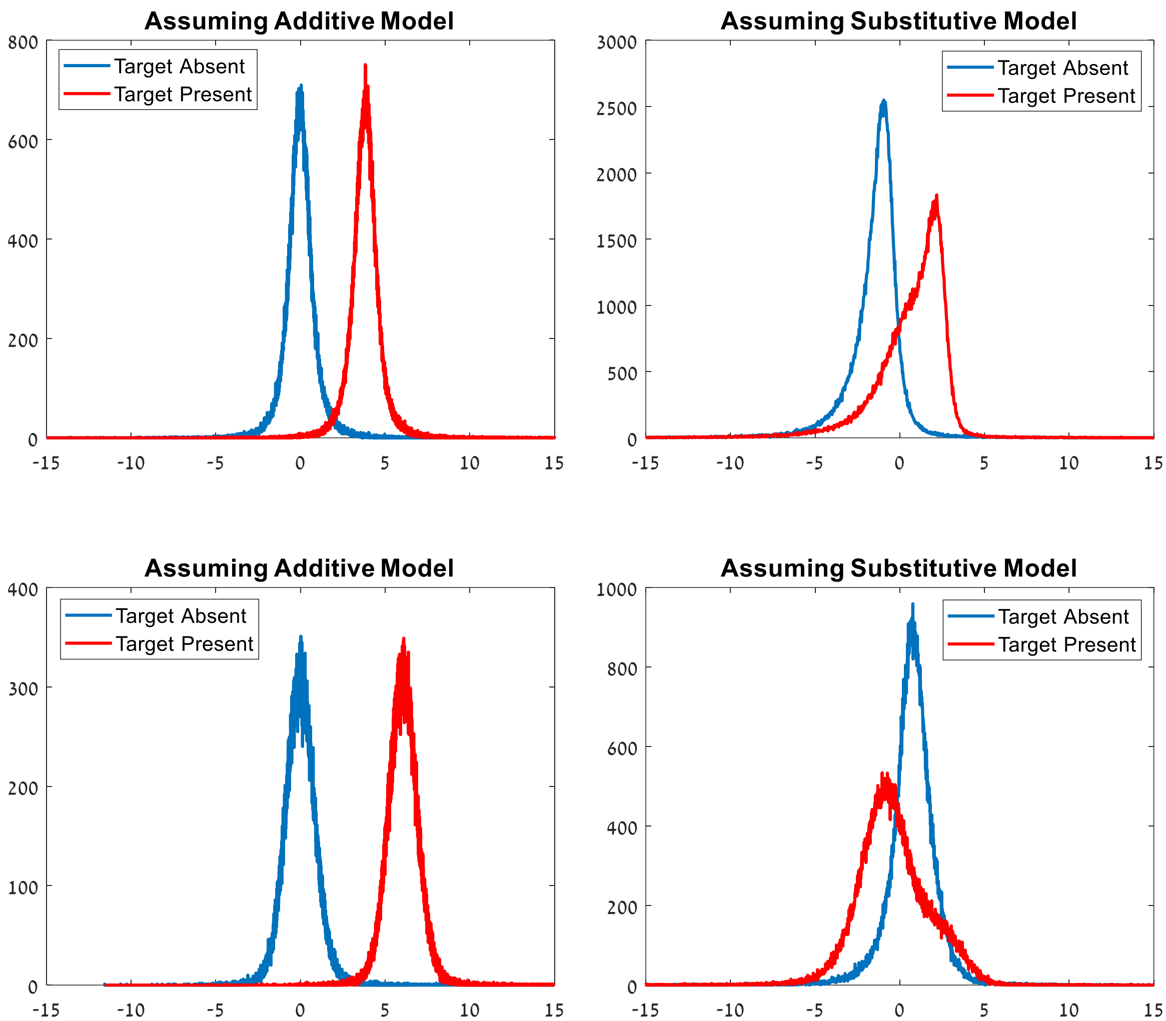


Figure 9: Detection Histograms while input is inaccurate (R=10), and target is substitutive (p=10%), for the RIT dataset (top) and for the Viareggio dataset (bottom)

The target model is substitutive, and therefore the expectation was that the substitutive AMF will show better separation between the target-present and the target-absent histograms. However, in this case, the additive AMF shows better results.

Figure 10 shows the ROC curves acquired for each image, assuming that the target's amplitude is inaccurate. The results are repeated for R values as low as 2, regardless of the fact that R=10 in figure 10.

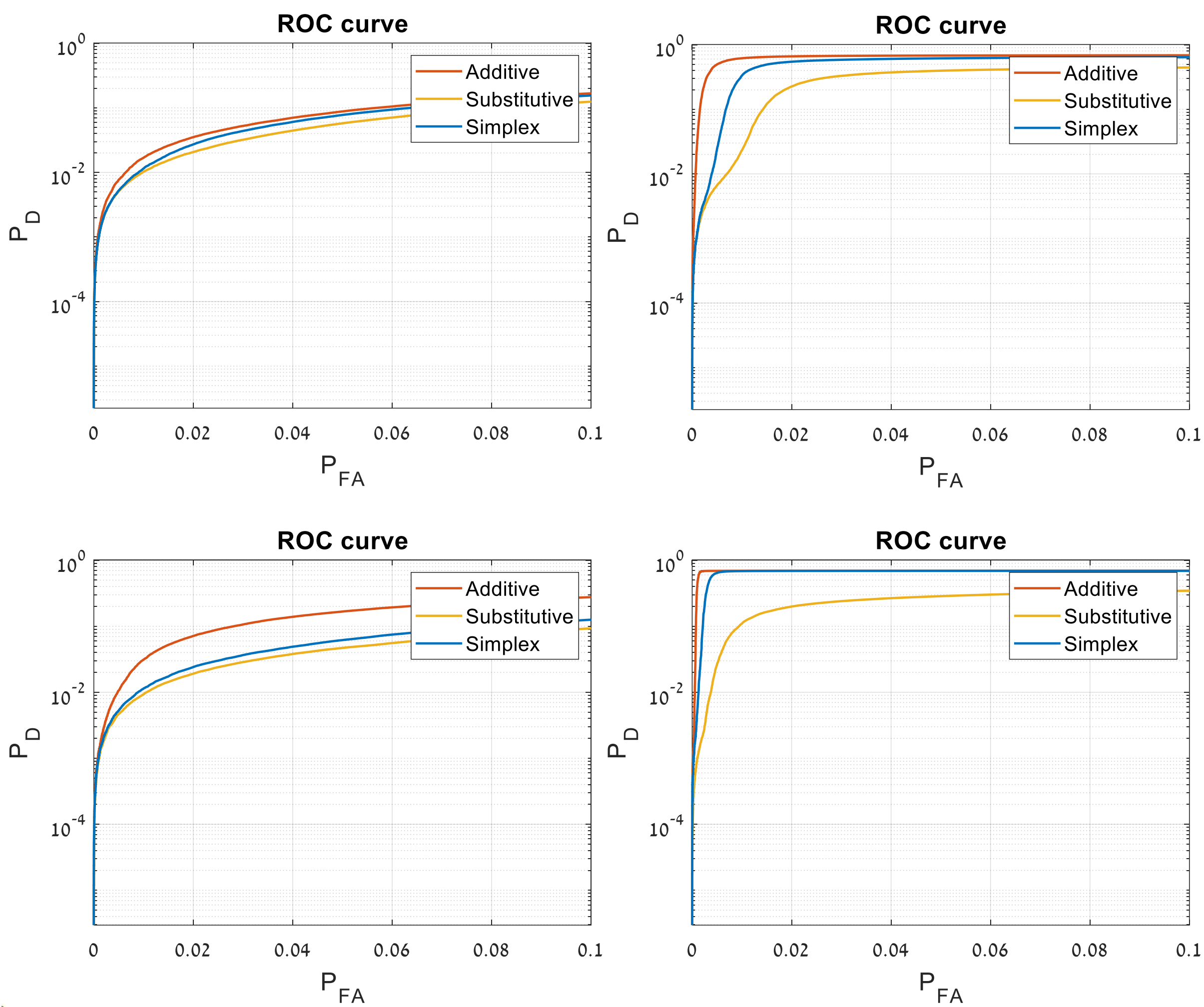


Figure 10: ROC curves for Detection when estimated amplitude of the target is inaccurate (R=2) for the RIT dataset with an additive target (top left), the RIT dataset with a substitutive target (top right), the Viareggio dataset with an additive target, and the Viareggio dataset with a substitutive target.

As opposed to the results seen in figure 10, in the case of inaccurate data the Additive AMF yields the best results regardless of the target model used. However, the Simplex AMF results remain between the other two algorithms' results, thus continuing to be a good compromise in case of an unknown target model. Furthermore, unlike in the case where the prior knowledge of the target was accurate, the results for an inaccurate target amplitude were consistent for all the target signatures we have used.

Table 2 shows the AUC calculations for detection where the target amplitude is inaccurate. As one can see, the AUC results for the Additive AMF are, for every target model, the highest ones. In addition, once again, the AUC results of the Simplex AMF are in-between.

Table 2: AUC calculations, for different scenarios, assuming inaccurate target amplitude

		RIT Dataset			Viareggio Dataset		
		Additive Detection	Substitutive Detection	Simplex Detection	Additive Detection	Substitutive Detection	Simplex Detection
Additive Target Model	Th=0.1	0.0439	0.0124	0.0326	0.1288	-0.0018	0.0155
	Th=0.01	0.0029	0.0001	0.0004	0.0076	-0.0003	0.0004
Substitutive Target Model	Th=0.1	0.9204	0.3702	0.7403	0.988	0.2585	0.9605
	Th=0.01	0.5290	0.0028	0.0872	0.8899	0.0343	0.6578

7. Conclusions

Of the six target signatures we examined, five gave unequivocal results, while the sixth target gave mixed results.

The conclusions we have drawn are:

1. Inaccurate estimation of the background of the pixel causes the Substitutive AMF to give a larger error than the Additive AMF.
2. For an unknown target's model or an inaccurate target's amplitude, the simplex algorithm constitutes a decent compromise.
3. For an inaccurate amplitude of the target, the additive AMF yields the best results, regardless of the target model.

8. Bibliography

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